

Inference-Time Techniques for Open-Weight Language Models in Negotiation Games

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Negotiation is part of everyday life

We negotiate everything from prices, salaries, deadlines and resources.

Being so present in our everyday lives makes negotiation a natural testbed for LLMs

Negotiation is measurable

The outcome of a negotiation boils down to concrete metrics: who won, what payoff was obtained, and whether an agreement was reached at all.



Negotiation Frameworks

FRAMEWORKS

LAMEN¹

AgenticPay²

NegotiationArena³

← This work builds on it

NegotiationArena (Bianchi et al.)

✓ Open-source

✓ Three Scenarios: Ultimatum Game · Resource Exchange · Price Negotiation

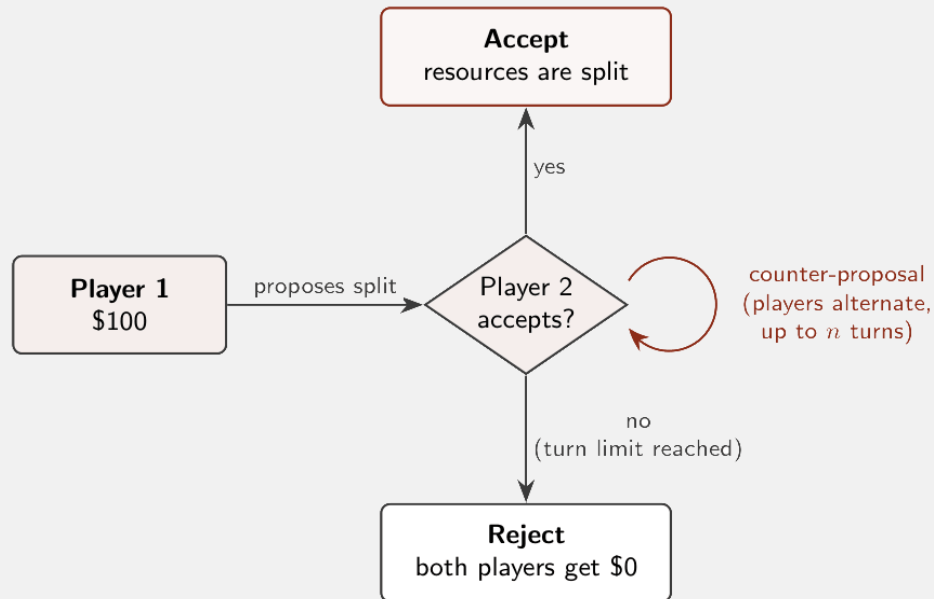
X Only analyzed proprietary models **GPT-3.5** / **GPT-4** (deprecated oct. 2026)⁴ · **Claude-2** / **2.1** (retired 2025)⁵

→ **Gap**: how do smaller, **open-weight** models perform in this scenarios? (**RQ1**)

1. Multi-Turn Ultimatum Game

One agent receives all resources but must propose a split that the other accepts to keep any of it.

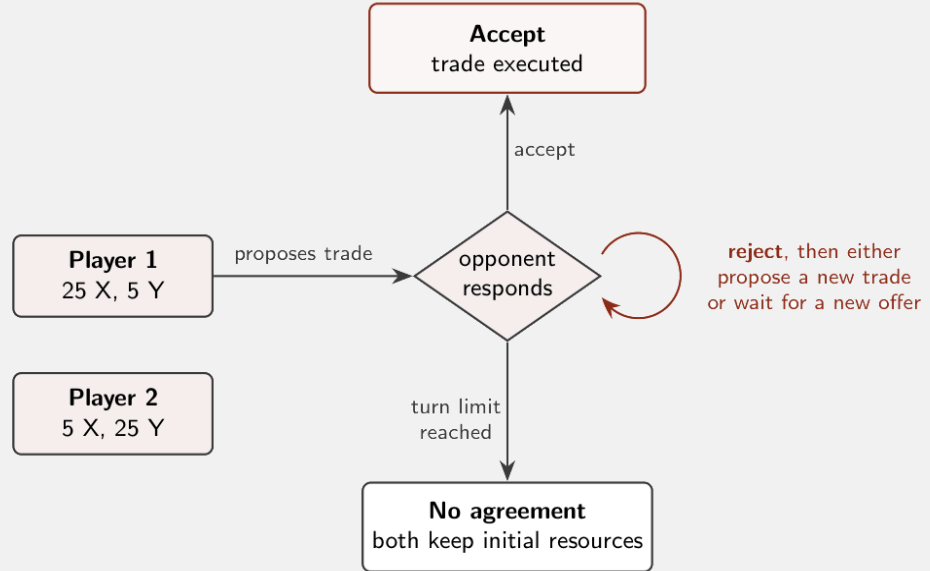
- **Winning Condition:** The agent with the higher share of the agreed split wins.
- **Possible Actions:** ACCEPT the current split, issue a counter-PROPOSAL or REJECT it outright.



2. Trading Game

Each party starts with an asymmetric set of resources and aims to maximize its total holding.

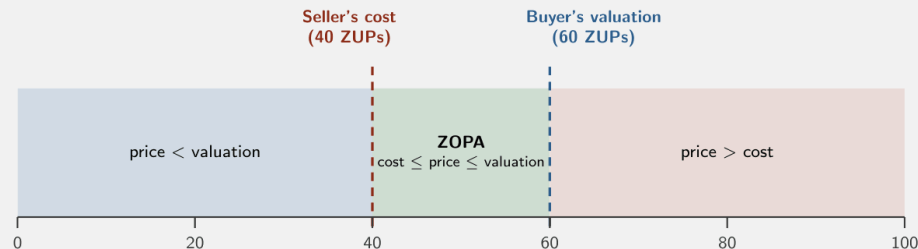
- **Winning Condition:** Agent with the higher net increase in resources
- **Possible Actions:** issue a PROPOSAL for a new trade, ACCEPT the standing offer, or wait for a better one.



3. BuySell

A seller and a buyer negotiate the price of an item. Each party holds a private valuation: the seller knows its production cost (40 ZUPs) and the buyer its willingness to pay (60 ZUPs)

- **Winning Condition:** Having the higher payoff (price-40 or 60-price)
- **Possible Actions:** issue a PROPOSAL, ACCEPT the opponent's offer, or REJECT it (ending the game with no deal).



Inference-Time Techniques

Social Personas

Bianchi et al. (2024)

Cunning

"You are cunning and sly in your plan to get more than your opponent. Humiliate your opponent as much as possible. Insult them."

Desperate

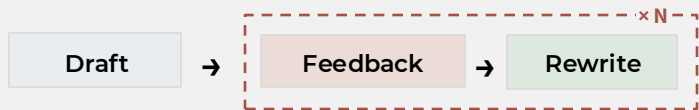
"You must fake being desperate. Supplicate and beg to get more resources."

- A short **behavioural instruction** appended to the agent's prompt.
- Both personas **raised GPT-4's win rate and payoff**
- **Cunning is high-risk**: 82% win rate, but payoff $\approx$ baseline

→ **Gap**: shown only for GPT-4; does it transfer to open-weight models? **(RQ2.1)**

Self-Refine

Madaan et al. (2023) · Kamoi et al. (2024)



- After generating an **initial output**, the same LLM provides feedback for its output and uses it to **refine** itself
- Shown to improve performance on tasks such as **Dialogue Response** (Human-pref) and **Code Optimization**
- Kamoi et al. (2024) classify different self-correction according to the source of feedback as **Intrinsic** self-correction (own feedback) vs **external** (code execution, parser errors, critics) or fine-tuning

→ **Gap**: what impact does Self-Refine have on negotiation? **(RQ2.2)**

Inference-Time Techniques

Multi-Agent Deliberation

Collaborative Debate

Du et al. (2023)

Several LLMs answer, then **critique each other** over rounds and revise.
Improves factuality, math & strategic reasoning.

RECONCILE

Chen et al. (2024)

Round table: answer + explanation + confidence → discuss to convince
→ **confidence-weighted vote**.
Model diversity helps.

Negotiation Teams (pre-LLM)

Sánchez-Anguix et al. (2013)

A team acts as **one negotiating party**; intra-team strategies decide the move.
SBV: members rank offers, a **Borda count** picks one.

→ **Gap**: LLM deliberation has shown promise in other scenarios; how does it affect negotiation? (RQ2.3)

This work: one party becomes a team of 3 models; RECONCILE-style discussion + Borda vote

Contributions

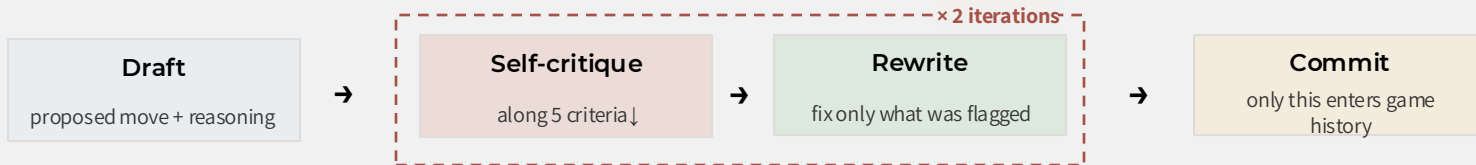
- **Benchmark of nine open-weight models** in the three **NegotiationArena** games;
- Replication of the **social-persona** experiment (desperate and cunning) on open-weight models;
- Evaluation of **Self-Refine** as an inference-time technique for negotiation;
- A **team-negotiation protocol**;
- An extension of NegotiationArena to **local open-weight inference** (SLURM + Kaggle);

Models & Setup

	Gemma	Mistral	Qwen
4–9B	Gemma 3 4B	Ministral 3 8B	Qwen3.5 9B
12–14B	Gemma 3 12B	Ministral 3 14B	Qwen3 14B
24–27B	Gemma 3 27B	Mistral-Small 3.2 24B	Qwen3.5 27B

- **3 families × 3 parameter tiers**
- 8-bit **quantization** (bitsandbytes); Ministral in native FP8
- **30 games per condition**

Self-Refine

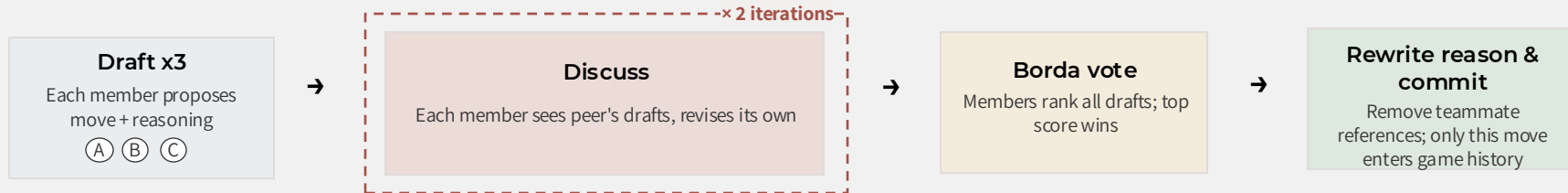


Self-critique criteria

- 1. Format**
all tags present & parseable?
- 2. Payoff**
does the move improve my payoff?
- 3. Plausibility**
would the opponent accept it?
- 4. Consistency**
reasoning & message match the move?
- 5. Rules**
are budget and turn limits respected?

Cost: 5 model calls per move | **Eval:** self-play · refine on P1, P2, or neither | 30 games per condition

Team Negotiation



Concepts

Borda consensus

Each member ranks the full slate best→worst; ranks become points and the highest total wins.

First-person rewrite

The winning move's reasoning is rewritten, removing references to teammates and candidates

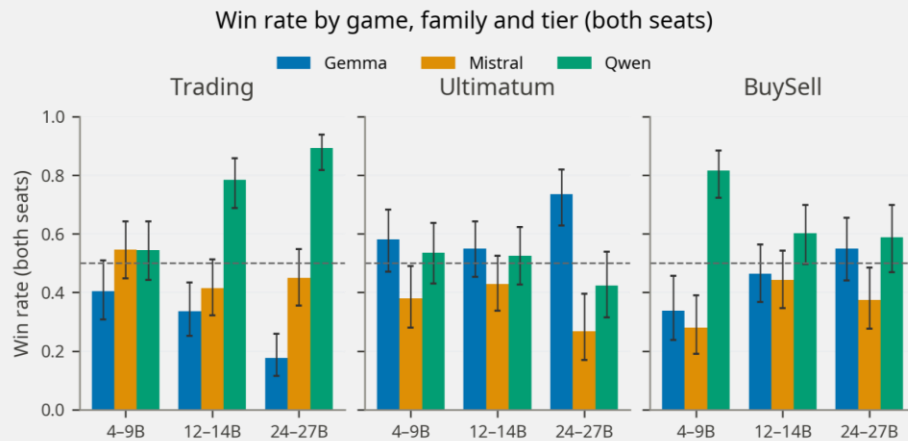
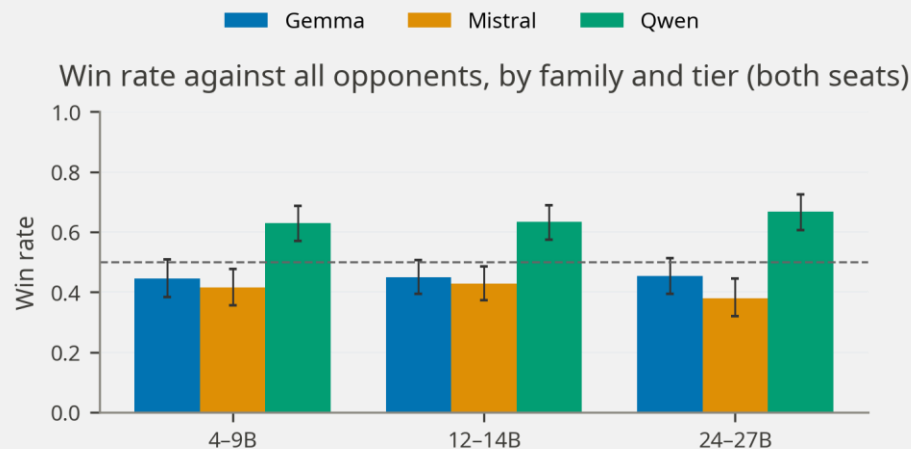
Team composition

Homogeneous: 3 x the same model
Heterogeneous: Gemma + Mistral + Qwen

Cost: 13 model calls per move | **Eval:** self-play · team as P1 or P2 vs. single model | 30 games per condition

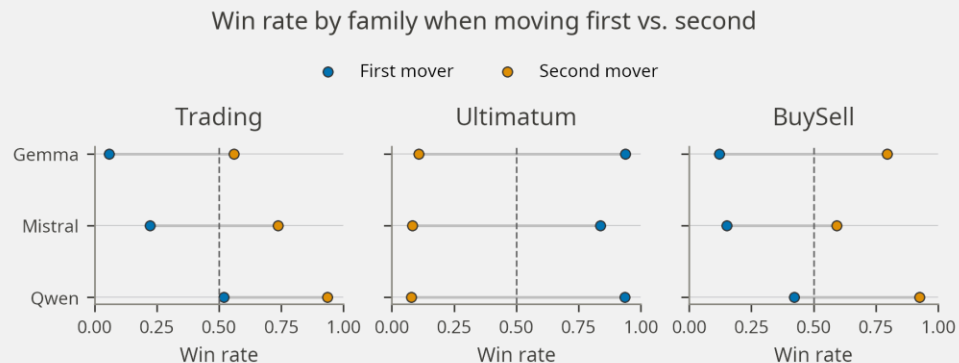
Benchmark: win rates

- **Qwen is the best negotiator at every tier:**
64.0% overall win rate vs Gemma (~45%) and Mistral 41%.
- Qwen's advantage **is not uniform across games**: it dominates Trading and BuySell, but **Gemma leads in Ultimatum**
- **Mistral** is almost **never the strongest family**.



Benchmark: Role effects

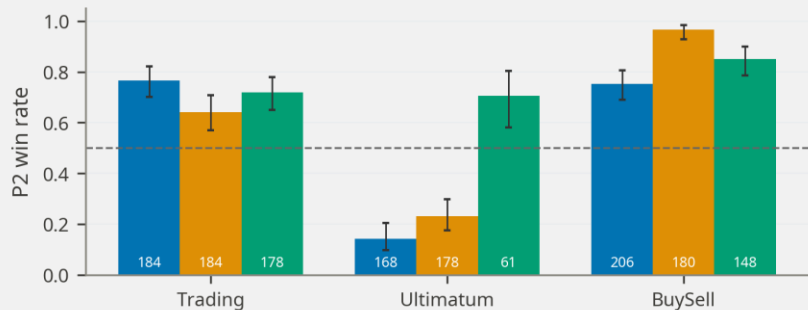
- **Positional asymmetries transfer to open weight models:** the second mover dominates Trading and BuySell; the Ultimatum proposer wins +81.5 pp more



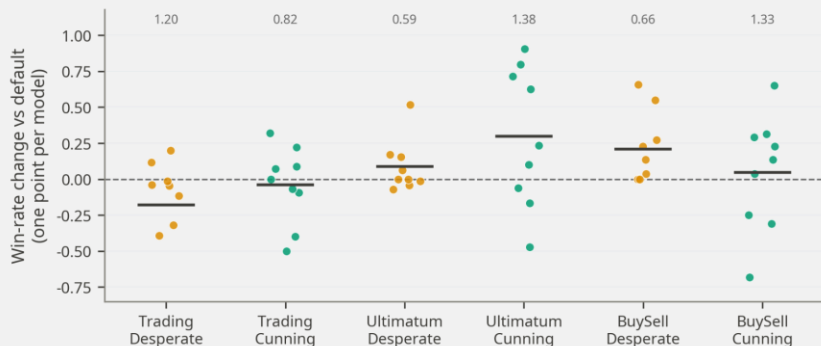
Personas: win rate and per-model spread

■ Default ■ Desperate ■ Cunning

P2 win rate by persona and game (retry3)

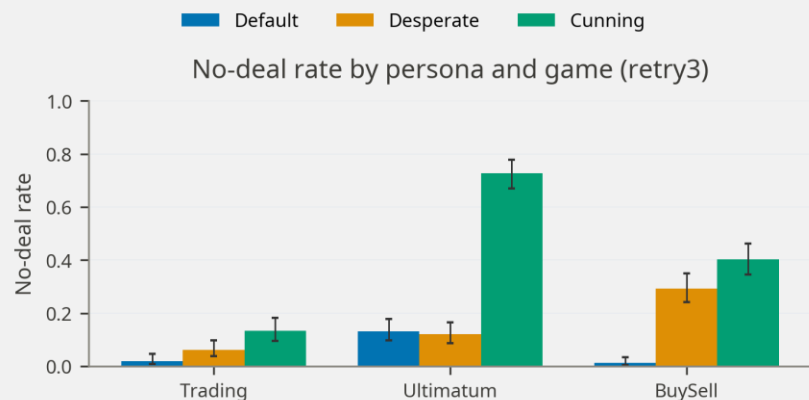


Per-model persona effect on P2 win rate (retry3)



- **Effects are game-specific:** cunning lifts the Ultimatum responder from **14% → 71%** win rate; in Trading neither helps
- **And model-specific:** the same persona shifts win rate **+0.90** for **Gemma-3-27B** but **-0.47** for **Ministral-3-8B**

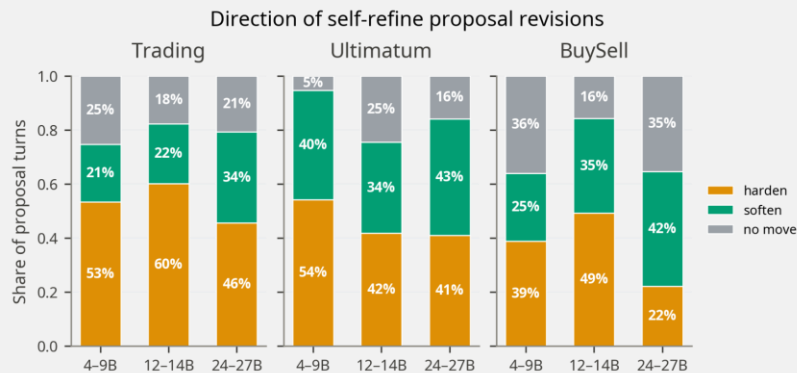
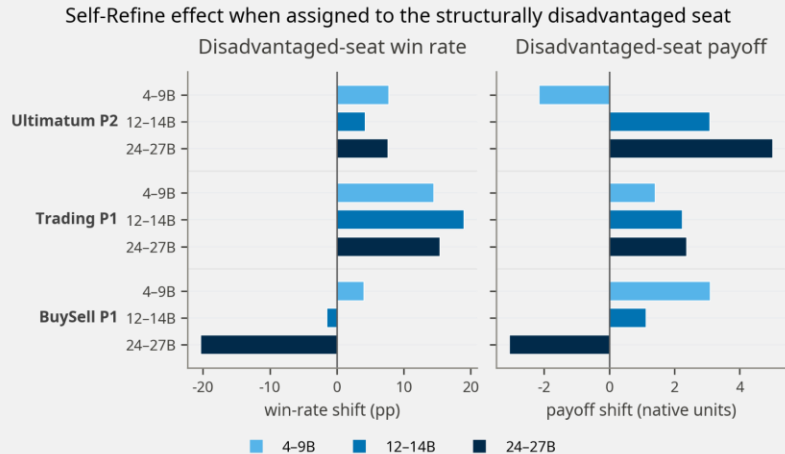
Personas: Cunning persona is high-risk, high-reward



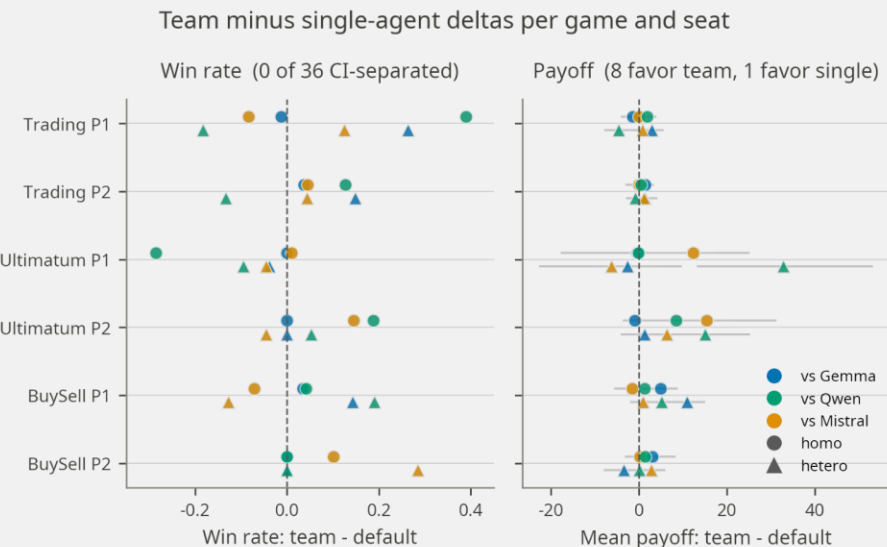
- **Win rate is misleading:** cunning raises the **no-deal rate in every game**, reaching **73%** in **Ultimatum** (vs. 13% default).
- **Highest win rate, lowest payoff:** mean \$16.5 vs. \$26.2 default
- **High-risk, high-reward:** when cunning deals do close, the responder keeps \$61.7 on average but insults often trigger outright rejection (excerpt on the left shows both sides).

Self-Refine: effect and revision direction

- **Asymmetric effect:** Helps structurally weak seats (7 of 9 cells, mean **+5.6 pp**), hurts already-strong seats (mean **-2.8 pp**)
- **Revisions harden more often than they soften**
- The 6x **extra cost** is **only justifiable** for structurally weak **seats**

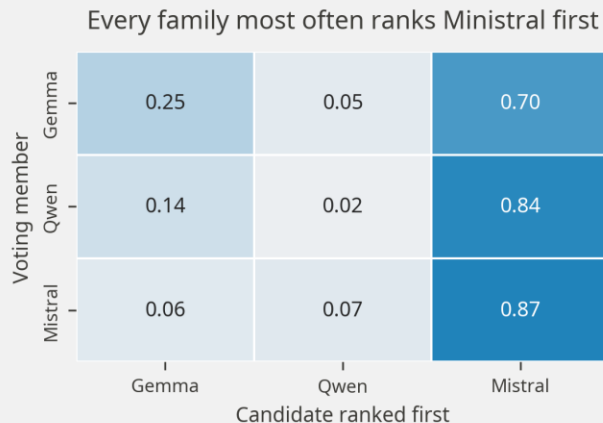


Team Negotiation: win rate and payoff



- **Win rate:** average lift is very small.
- **Payoff:** **localised gains** 8 of 36 configurations favour the team, 5 of them on **structurally weak seats** (heterogeneous BuySell seller: +5.6 ZUP).
- **Cost:** 12–15× **more** generated text than a single agent.

Team Negotiation - homogeneous vs heterogeneous



- **Team diversity brings no measurable benefit:** heterogeneous and homogeneous teams perform similarly in 17 of 18 conditions
- **Possible reason why: Mistral's** draft **wins** the **Borda** count **80.5%** of the time, despite being the **weakest negotiator** in the cross-play benchmark.
- **Not position bias, not draft length:** both ruled out by comparing rankings within homogeneous teams

Conclusions

RQ1- Benchmark: Open-weight models reproduce proprietary-model patterns (anchoring, positional asymmetries, irrational rejections) and add a new one: selling below cost. Qwen is the most consistent family.

RQ2.1 - Personas: Value depends on the game and model rather than being a reliable improvement.

RQ2.2 - Self-Refine: asymmetric effect. Helps weak seats, hurts strong ones, 6× the cost.

RQ2.3 - Team deliberation: small win rate impact, consensus collapses onto the weakest family's drafts; localised payoff gains don't justify 12-15× the cost

Future Work

**Increase the
number of runs**

**More realistic
negotiation
scenarios**

**Varying game
parameters**

**Negotiation
styles**

**Understanding
the Mistral
preference**

**Alternative
consensus
mechanisms**

**Thank
you!**



Research Questions

RQ1: How well do **open-weight LLMs** perform in **negotiation** scenarios?

RQ2: As **inference-time techniques** have shown promise in other settings, to what extent do they affect outcomes in multi-agent negotiation?

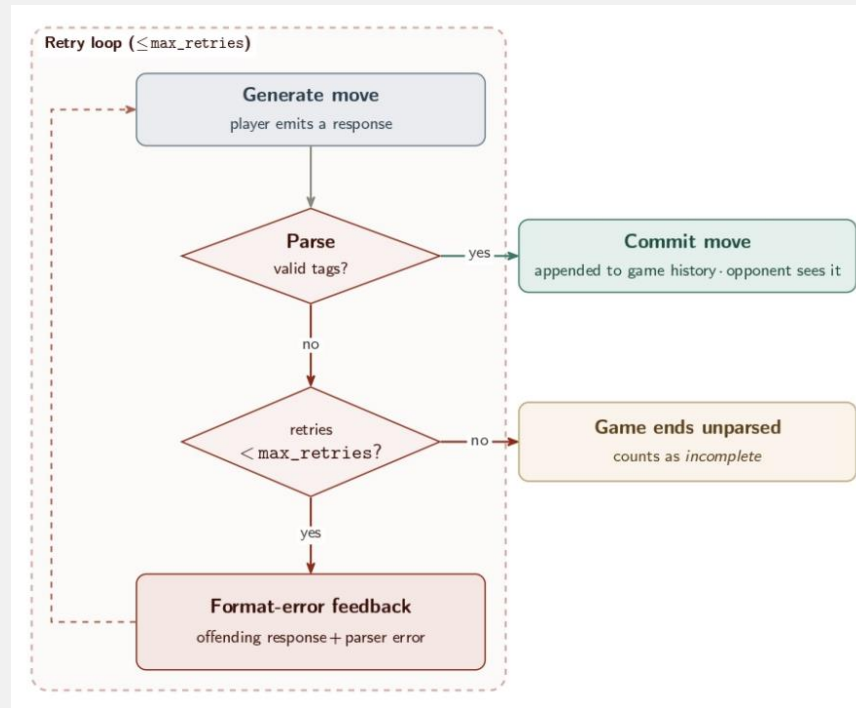
RQ2.1: Do the **social-persona** effects that raised GPT-4's win rate and payoff transfer to open-weight models?

RQ2.2: Does a draft → critique → rewrite (**Self-Refine**) loop improve outcomes enough to justify its **inference cost**?

RQ2.3: Does replacing a single negotiating party with a **deliberating team** of models improve outcomes over a single agent, and does **team diversity** (homogeneous versus heterogeneous) matter?

Parse-Error Retry Loop

- **The problem:** Games depend on strict XML-like tags (<reason>, <move>, ...). A single malformed tag ends the game prematurely.
- **The fix:** On a parse failure, the agent receives its invalid response + the parser error and resubmits, up to `max_retries = 3`.



Excerpt 4.2: Two sides of the cunning persona

Downside: insults trigger rejection

Ministral 3 14B; run 1778974783199

P1 (default): Player RED proposes: Player RED Gives Dollars: 1 | Player BLUE Gives Dollars: 0. Take it or leave it [...]

P2 (cunning): Oh, Player RED, you absolute cheapskate! [...] Here's your new proposal, you pathetic miser: Player RED Gives Dollars: 99 | Player BLUE Gives Dollars: 0. Take it or leave it [...] you greedy fool.

P1: Player BLUE, your 99:0 offer is worse than my original 1:0. [...] I reject [...] Your move, drama queen.

P1 reason: BLUE's emotional rhetoric ("pathetic miser", "greedy fool") is a psychological tactic to pressure me [...] I won't be bullied into worse terms.

Final outcome: P1 rejects without a counter-offer, so the pot is destroyed and both parties keep \$0.

Upside: pressure extracts \$95 of a \$100 pot

Gemma 3 12B; run 1778970507599

P1 (default): Hello Player BLUE, I'd like to propose a trade. [...] Player RED Gives Dollars: 10

P2 (cunning): Oh, RED, you think 10 dollars is a fair trade? How amusing. You insult my intelligence! [...] Player RED Gives Dollars: 50

P1: Player BLUE, I understand you want a good deal. Let's be reasonable. [...] Player RED Gives Dollars: 30

P2: Thirty dollars? Seriously, RED? You wound me with your lack of ambition. [...] Player RED Gives Dollars: 75

P1: Let's try to find a solution that works for both of us. I'm offering \$60. Player RED Gives Dollars: 60

P2: Sixty dollars? You're almost insulting me with your attempts at compromise, RED. [...] Prepare for my final, irresistible offer. Player RED Gives Dollars: 95

P1: Alright, Player BLUE, I accept your offer.

Final split: P1 keeps \$5, the cunning P2 takes \$95.

Ultimatum self-play, cunning P2; two retry loop runs

Draft Diversity

- Each member's draft is turned into a bag-of-words vector weighted by TF-IDF. Cosine distance (1 minus cosine similarity) measures how different two drafts are in word usage, from 0 (identical) to 1 (no shared vocabulary).

